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Paper Details

Paper Code: CS2580

Paper Title: Database Systems

Exam Period: April 2024

Exam Rubric

Time Allowed: 2 hours

Exam Type: Standard Examination

Approved Calculators: Permitted

Instructions

Answer **FOUR** questions.

If more than **FOUR** questions are answered, please clearly indicate which questions you wish to be marked. Otherwise, the first **FOUR** questions answered in the answer booklet will be marked.

Read carefully the instructions on the answer booklet and make sure that the particulars required are entered on each answer booklet.

The exam question paper **MUST NOT** be removed from the examination venue.

Permitted calculators: CASIO FX82, FX83 and FX85 series

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Question 1 [25 marks]

a) Consider the following tables and queries:

Student

StudentID	StudentName	Age	Avg_Mark
1234567	Peter	18	65
1224467	Alice	19	70
1244467	Jane	20	75

Registered

StId	CourseID	Mark
1234567	CS258	65
1244467	CS346	80
1244467	CS260	68
1224467	CS258	67
1234567	CS346	65

Query1: SELECT StudentName
FROM Registered LEFT OUTER JOIN Student ON StudentId = StId;

Query2: SELECT StudentName
FROM Student LEFT OUTER JOIN Registered ON StudentId = StId;

Will Query1 and/or Query2 produce any errors? If yes, explain briefly. If not, will they give the same results, or different results?

[3]

b) Can you describe a scenario (i.e. a table and its attributes) where a foreign key would need to refer to an attribute in the same table?

[3]

c) Consider the following table.

A	B
1	2
3	2
5	1
6	1
2	6
9	5
10	2

i. Is it possible to identify which of the columns "A" or "B" given in the table can be the "primary key" or a "foreign key"? If yes, do so and explain. If not, explain.

[3]

ii. Suppose that during the creation of the above table it was specified that **'ON DELETE CASCADE'** and that the row (A=1, B=2) is deleted from the table. Would any more tuples need to be deleted? If yes, which? If not, explain.

[4]

d) You are given a table, "EMPLOYEE," which has two columns ('Name' and 'Salary'). The Salary column in this table has some NULL values. Suppose we want to find the records which have NULL values. Consider the output for the following queries?

Query1. SELECT * FROM EMPLOYEE WHERE Salary <> NULL;
Query2. SELECT * FROM EMPLOYEE WHERE Salary = NULL;

Can we say what the output of these queries will be? If not, explain with one sentence. If yes, show the outputs of Query1 and of Query2.

[2]

e) Two tables, T1(ID, Age, Name) and T2(ID, Age, Name) have no NULL values and have 100 records each. Consider the following two queries:

```
Query1:  SELECT T1.ID FROM T1
         WHERE T1.Age > ALL
               (SELECT T2.Age FROM T2 WHERE T2.name = 'Josh')
Query2:  SELECT T1.ID FROM T1
         WHERE T1.Age > ANY
               (SELECT T2.Age FROM T2 WHERE T2.name = 'Josh')
```

Can you say with certainty whether Query1 will give a larger or equal number of rows compared to Query2, or whether Query1 will give an equal number of rows compared to Query 2, or whether Query1 will give a smaller or equal number of rows compared to Query2? If not, explain why. If yes, explain your answer.

[2]

f) Is the "SELECT" operation in SQL equivalent (in the sense of producing the same output) to the "PROJECT" operation in relational algebra?

[3]

g) Consider the relation S(A, B) in which (A, B) is the primary key and the relation T(A, C) where A is the primary key. Assume there are no null values and no foreign keys or other integrity constraints.

Query1: select A from S where A in (select A from T)

Query2: select A from T where A in (select A from S)

Will the two queries give the same result, different results, or sometimes the same and sometimes different result? Briefly explain which case(s) is (are) possible and when.

[5]

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Question 2 [25 marks]

Consider a database for hotel bookings:

HOTEL (hotelNo, hotelName, city)
ROOM (roomNo, hotelNo, type, price)
BOOKING (hotelNo, guestNo, dateFrom, dateTo, roomNo)
GUEST (guestNo, guestName, guestAddress)

You must use the aliases H, R, B, G, respectively, for the table names in the queries below. Assume that guestAddress is of type VARCHAR[50], containing the city name. Similarly, guestName is of type VARCHAR[30], containing the first name.

Specify the following queries in SQL:

a) List the names and addresses of all guests living in London, with a booking in a hotel in 'London', alphabetically ordered by name. You must use explicit joins (i.e., using the correct 'JOIN' operator).

[4]

b) List the guest numbers associated with the first name 'Jane' having made bookings with unknown dateTo, using implicit joins (i.e., without explicit 'JOIN' operators).

[4]

c) Repeat (b) above but without using any explicit or implicit JOINs.

[8]

d) List the price and room number of rooms at the Hilton Hotel. Additionally, if the room is currently occupied list also the name of the guest staying in the room. You may use the CURDATE() function and assume that all dates are strings of the form yyyymmdd.

[9]

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Question 3 [25 marks]

a) Show the full proof of the derived 'union' rule using only Armstrong's Reflexive, Augmentation and Transitive inference rules.

[4]

b) Consider the following relation: R(A,B,C,D,E)

The following functional dependencies are known to hold:
FD1. {B,C}->{D}
FD2. {A}->{B}
FD3. {C}->{E}

i. Formally derive the candidate key(s) of this relation, using only Armstrong's Reflexive, Augmentation, Transitive and Union rules.

[8]

ii. Explain the highest possible normal form of R, stating any assumptions.

[2]

iii. Decompose R such that the new database is in the highest normal form possible and demonstrate what normal form your new design is in.

[4]

c) Given the relations S(A,B,C,D) and the functional dependencies:
{A,B}->{C,D}
B->C
C->D

Explain all necessary considerations for decomposing S into two separate tables. Hence, explain whether S1(A,B,C), S2(A,B,D) would provide all the benefits of the original table S.

[7]

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Question 4 [25 marks]

A relational database contains information about bank accounts.

Accounts(aID, cID, balance, type)

InterestRates(type, interest)

Customers(cID, cName, job, city)

Transactions(fAID, tAID, amount, time)

- Accounts
  - aID: account ID
  - cID: customer ID
  - balance: amount in the account (e.g. 100 for £100)
  - type: account type (e.g. “Regular Saver” or “Current Account”)
- InterestRates
  - type: account type (e.g. “Regular Saver” or “Current Account”)
  - interest: number representing the interest rate (e.g., 2.4 for 2.4%)
- Customers
  - cID: customer ID
  - cName: the customer’s name
  - job: the customer’s job title
  - city: the customer’s city
- Transactions
  - fAID: account the money was paid from
  - tAID: account the money was paid to
  - amount: amount of money in the transaction
  - time: timestamp represented as an integer number of seconds since 1st January 1970.

You must use the aliases A, I, C, T, respectively, in the queries below.

a) Write a Relational Algebra expression to show the name of all customers who have an account that has a balance of at least 1000 and an interest rate of more than 5. Your answer must not use set operations. You may use variables.

[6]

b) Write a Relational Algebra expression to identify the names of pairs of customers who have transferred money to each other (i.e., the names of all customers x and y where customer x has ever paid customer y and customer y has ever paid customer x). Exclude any cases where customer x and customer y are the same customer. Ensure that each customer pair is only listed once. You may use variables but should not use set operations.

[8]

c) Write a tuple relational calculus query to retrieve the name and job title of customers who own at least one account for which there has never been a single withdrawal that has exceeded 500.

[6]

d) Explain what this query would return:

{ k | (exists a,b,c,d,e,f,g,h,i,j,l)(Transactions(a,b,c,d) and Transactions(e,f,g,h) AND b = e AND h < d AND Accounts(i,j,k,l) AND i = b) }

[5]

Question 5 [25 marks]

A company uses the following table to store information about their employees.

EMPLOYEE(SSN, Name, JobTitle, Location, JobPerformance, Salary)

Administrative staff should be able to see all attributes except JobPerformance.

Managers should be able to see all attributes.

All other employees should only be able to see Name, JobTitle and Location.

a) Explain how the above security policies could be enforced using a Relatational Database Management System such as Postgres. Describe how permissions could easily be updated if an employee was promoted to a Manager and provide samples of code that could be used to define this within SQL.

[6]

b) A larger company has similar security requirements to those shown above. However, the managers need to be further categorised into different management tiers; Top-level, Middle-level, Lower-level. In some circumstances, managers have different information about JobPerformance of a particular employee. Explain a technique that would allow different classifications of managers to record different information about the same employee and describe how this prevents confidential information from being inferred by a user with a lower classification.

[8]

c) The company now employs a group of external consultants to provide advice about whether their employees are being paid appropriately. The manager creates a web application that allows the consultants to see the aggregated salary information amongst different groups of employees (chosen by the consultants as any combination of the attributes JobTitle, Location, JobPerformance). Explain why this functionality may cause security problems, and briefly outline techniques that could be used to solve this.

[7]

d) Explain why prepared statements are important and briefly outline how they work.

[4]

End